Laplace Transform Notes

FRORDO

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1. What's the Point?

Differential equations are annoying to solve. Piecewise/discontinuous functions are also annoying. The Laplace transform is here to save the day!

2. Definition

Definitions

The *Laplace transform* is an operation that takes a function of a real variable (usually t) as input and outputs a function of a complex variable s. If $f(t)$ is our original function, $F(s)$ is its transform.

$$\mathcal{L}\{f(t)\}(s) = F(s) \stackrel{\text{def}}{=} \int_0^\infty f(t)e^{-st}dt$$

There is a slight caveat. The above integral may not converge. We will ignore this for now.

Example 1

Calculate $\mathcal{L}\{e^{at}\}$.

$$\begin{aligned}\mathcal{L}\{e^{at}\} &= \int_0^\infty e^{at}e^{-st}dt \\ &= \frac{1}{s-a}\end{aligned}$$

This is not exactly correct. The integral converges only when $\mathfrak{R}(s) > a$. But we will wave our hands and simply say

$$\mathcal{L}\{e^{at}\}(s) = \frac{1}{s-a} \text{ for } s \neq a$$

3. Properties of the Laplace Transform

3.1. Linearity

Linearity

The Laplace transform is linear!

$$\mathcal{L}\{af + bg\} = a\mathcal{L}\{f\} + b\mathcal{L}\{g\}$$

which follows from the linearity of integrals.

3.2. Existence

Existence Theorem

A function f is of *exponential order* if $|f(t)| \leq Me^{ct}$ for some positive M and c.

If f is continuous and of exponential order with constant c, then $\mathcal{L}\{f\}$ exists for all $s > c$. Proof follows by comparison.

3.3. Unique and Invertible

Existence Theorem

Two integrable functions have the same Laplace transform if they differ on a set of Lebesgue measure zero. For example, if two functions are different at only finitely many points, they have the same Laplace transform.

We will define an inverse operation

$$\mathcal{L}^{-1}\{F(s)\} = f(t)$$

4. Some Useful Laplace Transforms

Table of transforms

Assume f is sufficiently nice.

$$\begin{aligned}
\mathcal{L}\{e^{at}\} &= \frac{1}{s-a} \text{ for } s \neq a \\
\mathcal{L}\{\sin at\} &= \frac{a}{s^2+a^2} \\
\mathcal{L}\{\cos at\} &= \frac{s}{s^2+a^2} \\
\mathcal{L}\{t^n\} &= \frac{\Gamma(n+1)}{s^{n+1}} \\
\mathcal{L}\{u(t-a)\} &= \frac{e^{-as}}{s} \text{ where } u \text{ is the Heaviside step function} \\
\mathcal{L}\{\delta(t-a)\} &= e^{-as} \\
\mathcal{L}\{u(t-a)f(t-a)\} &= e^{-as}F(s) \\
\mathcal{L}\{f(t)\} &= \frac{1}{1-e^{-sp}} \int_0^p e^{-s\tau} f(\tau) d\tau \text{ where } p \text{ is the period} \\
\mathcal{L}\{f'(t)\} &= sF(s) - f(0) \\
\mathcal{L}\{f''(t)\} &= s^2F(s) - sf(0) - f'(0) \\
\mathcal{L}\left\{f^{(n)}(t)\right\} &= s^nF(s) - s^{n-1}f(0) - \cdots - sf^{(n-2)}(0) - f^{(n-1)}(0) \\
\mathcal{L}\left\{\int_0^\tau f(\tau) d\tau\right\} &= \frac{F(s)}{s} \\
\mathcal{L}\{e^{-at}f(t)\} &= F(s-a) \text{ (the shift property)} \\
\mathcal{L}\{f * g\} &= F(s)G(s)
\end{aligned}$$

5. Misc Notes

$$\mathcal{L}\{g\} \to \text{use by parts}$$

$$\mathcal{L}\left\{\int_0^t\right\} \to \text{change order of integration, otherwise convulution}$$

$$(f * g)(t) \overset{\text{def}}{=} \int_{-\infty}^{\infty} f(\tau) g(t-\tau) d\tau$$

$$\mathcal{L}\{f * g\} = F(s)G(s)$$

Set $G = 1$ (the step function) and the result follows.

This holds for nested integrals!

$$\mathcal{L}\left\{\int^n\right\} = \frac{F(s)}{s^n}$$

. This is true by Cauchy’s repeated integration formula, and then convolution theorem.